

PROYECTO DATALOGGER



ESCUELA TÉCNICA SUPERIOR DE INGENIEROS DE TELECOMUNICACIÓN

UNIVERSIDAD DE VALLADOLID

MÁSTER EN INGENIERÍA DE TELECOMUNICACIÓN

TALLER DE PROYECTOS 1: SISTEMAS ELECTRÓNICOS

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RESUMEN

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1.Introducción

2.Planificación

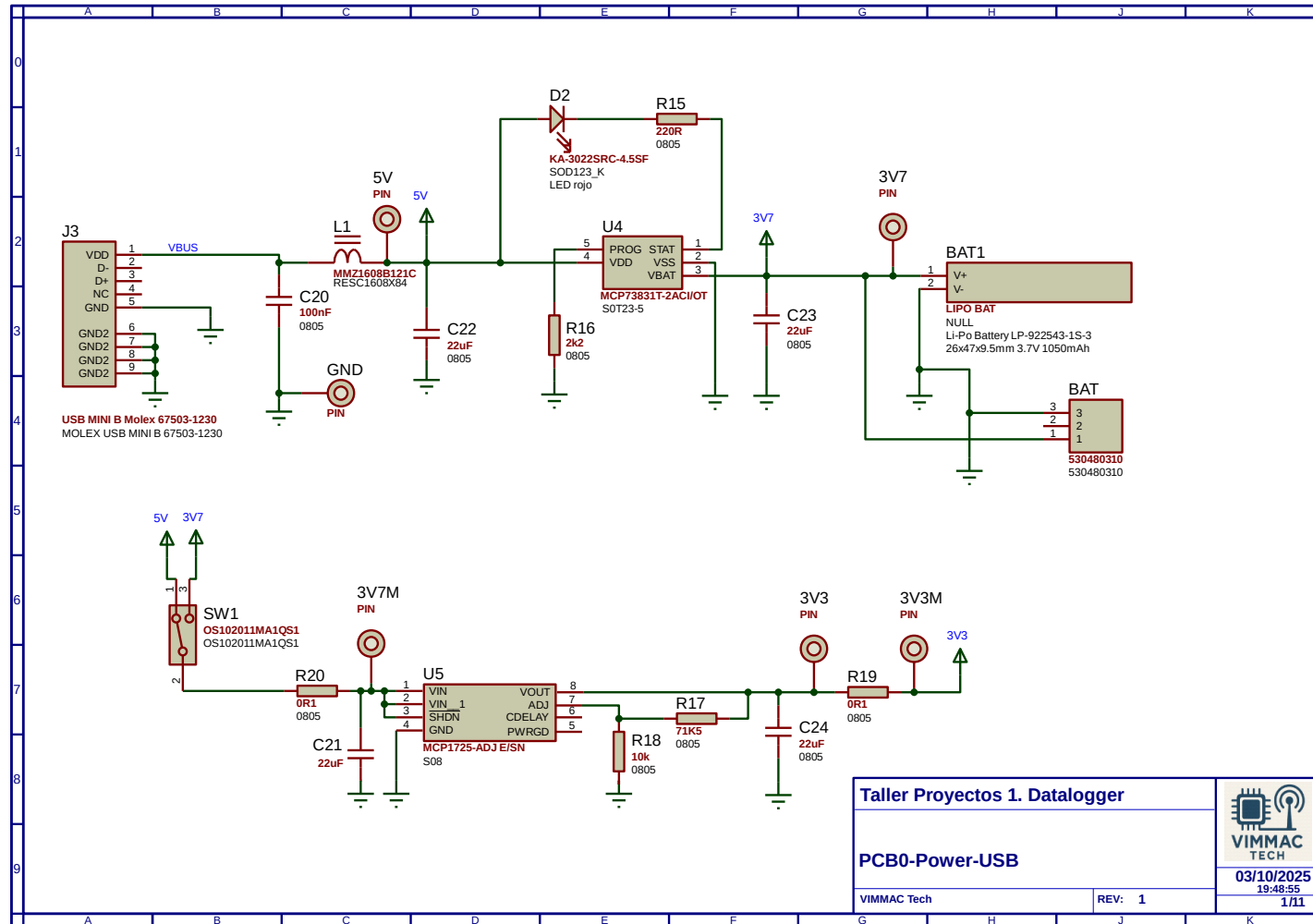
2.1. Diagramas de Gantt

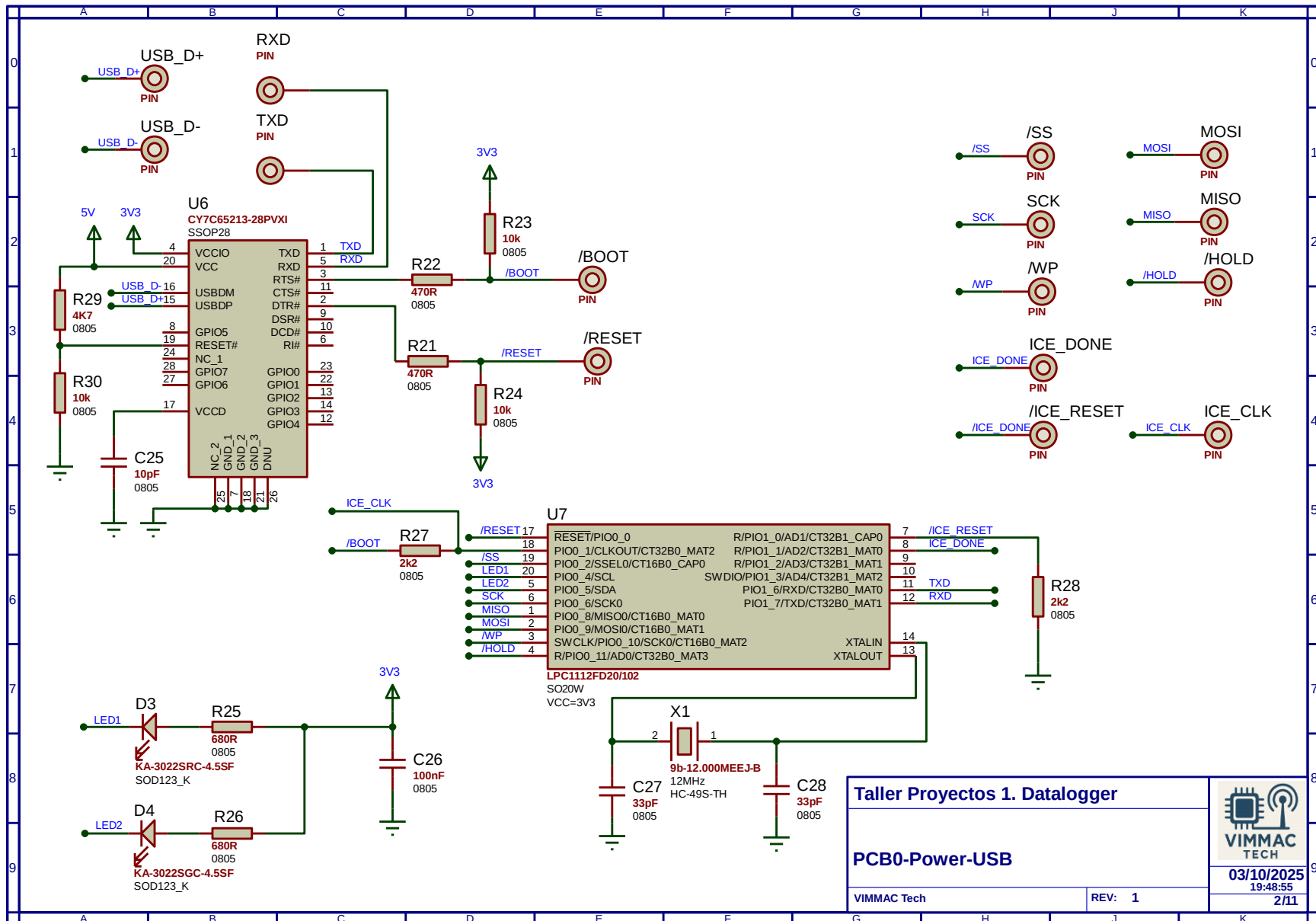
3.Hardware

3.1. Esquemas Electrónicos

3.2. Análisis eléctrico y térmico

ANEXO 1: ESQUEMATICOS





Taller Proyectos 1. Datalogger

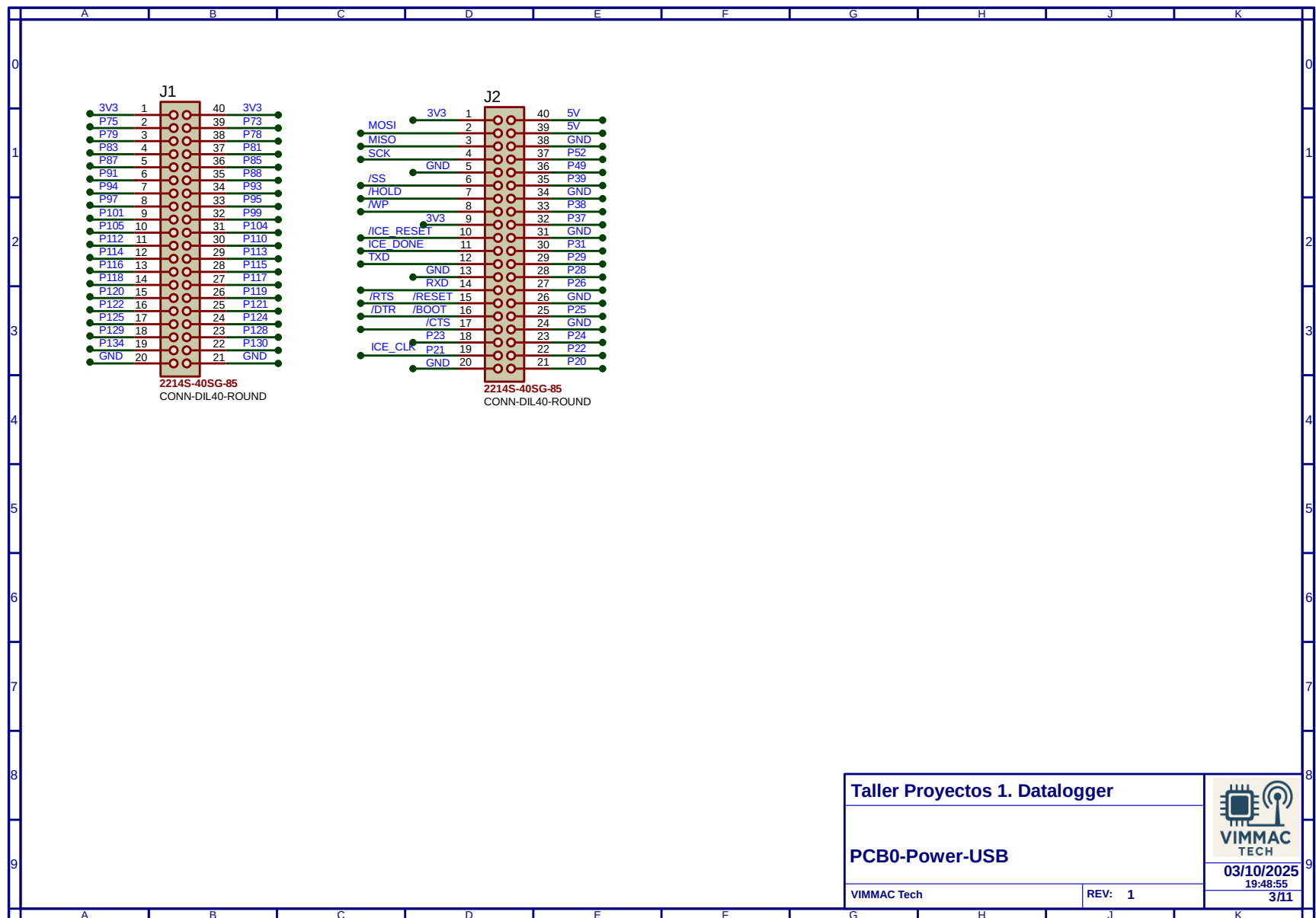
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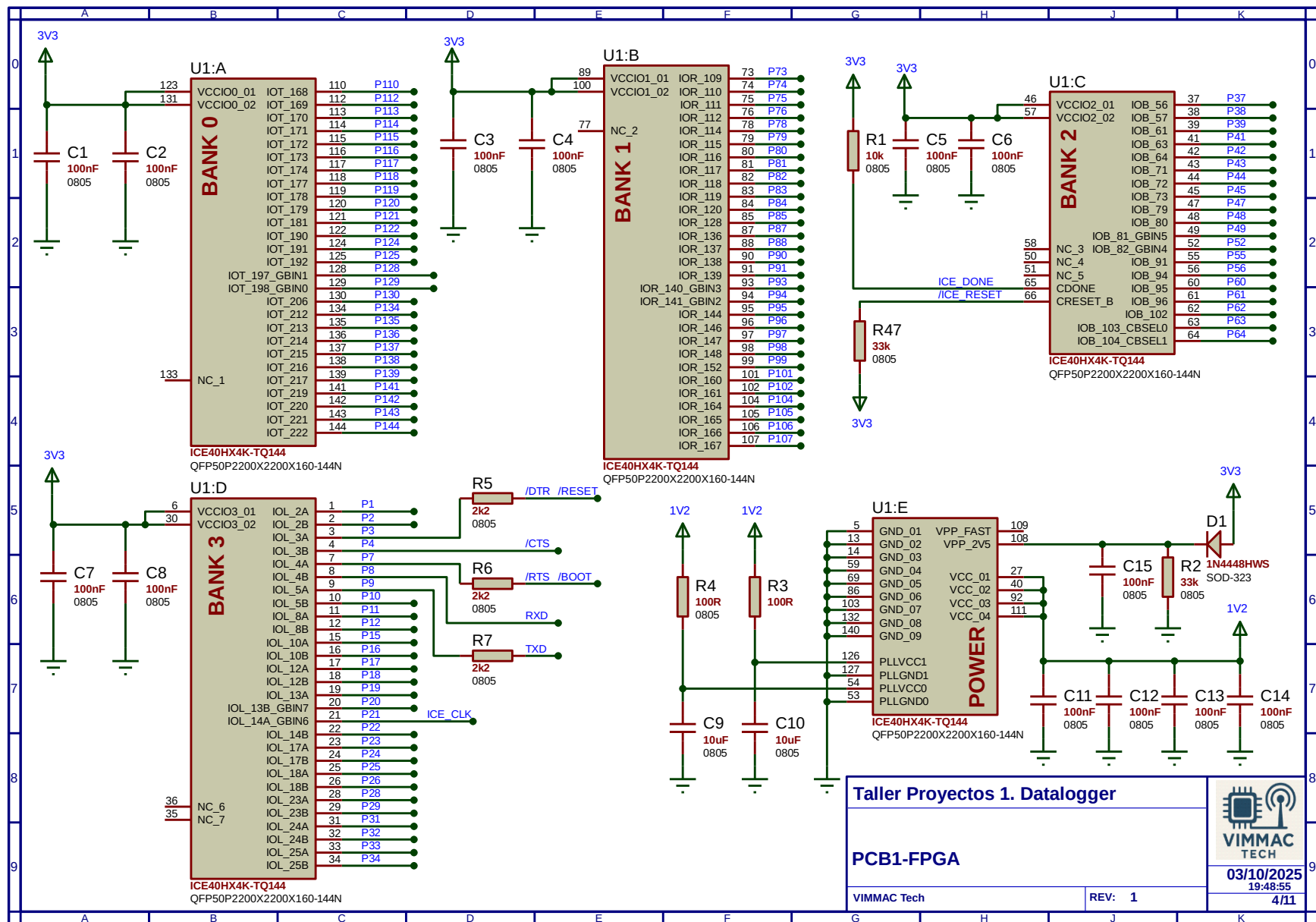
VIMMAC Tech

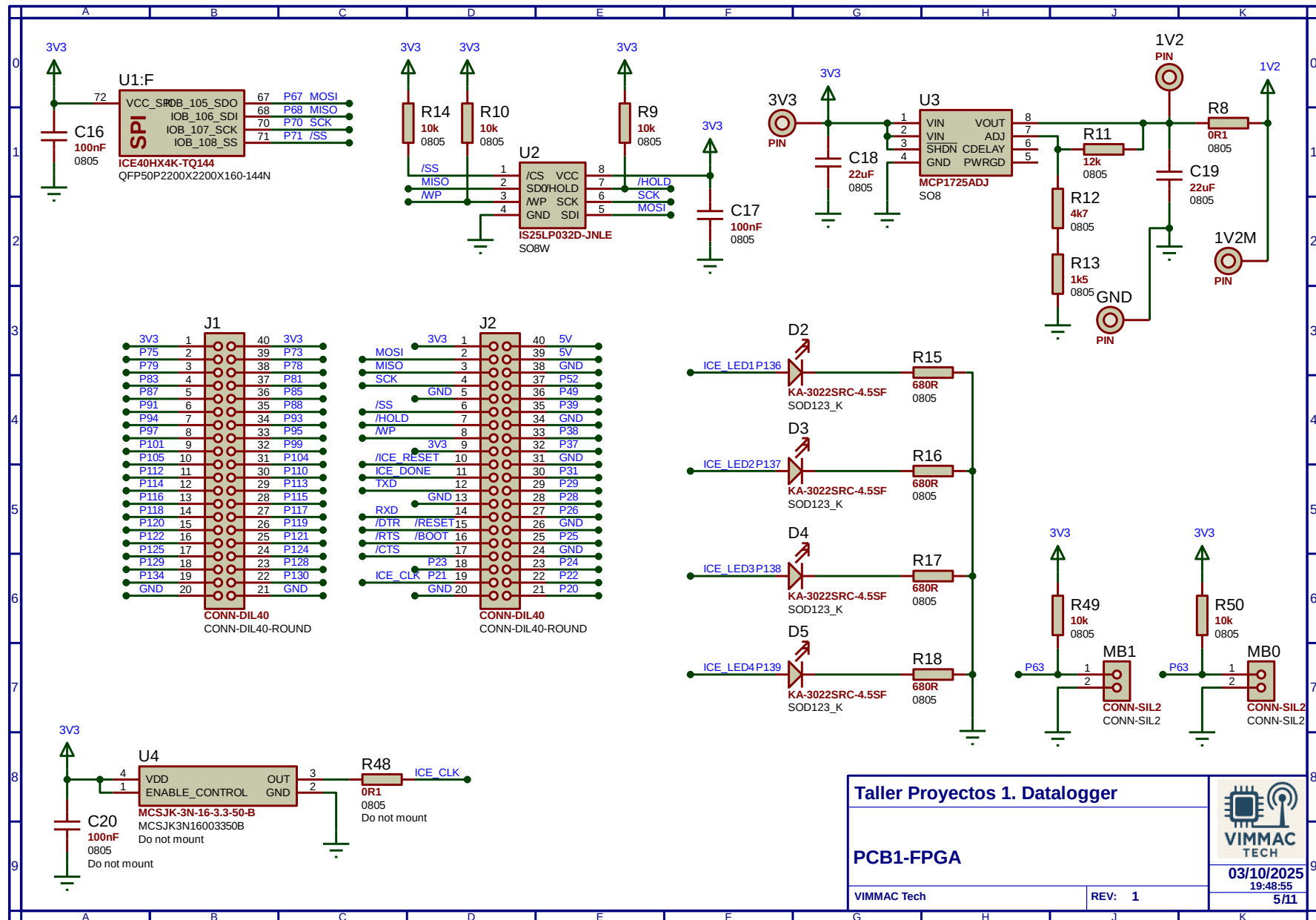
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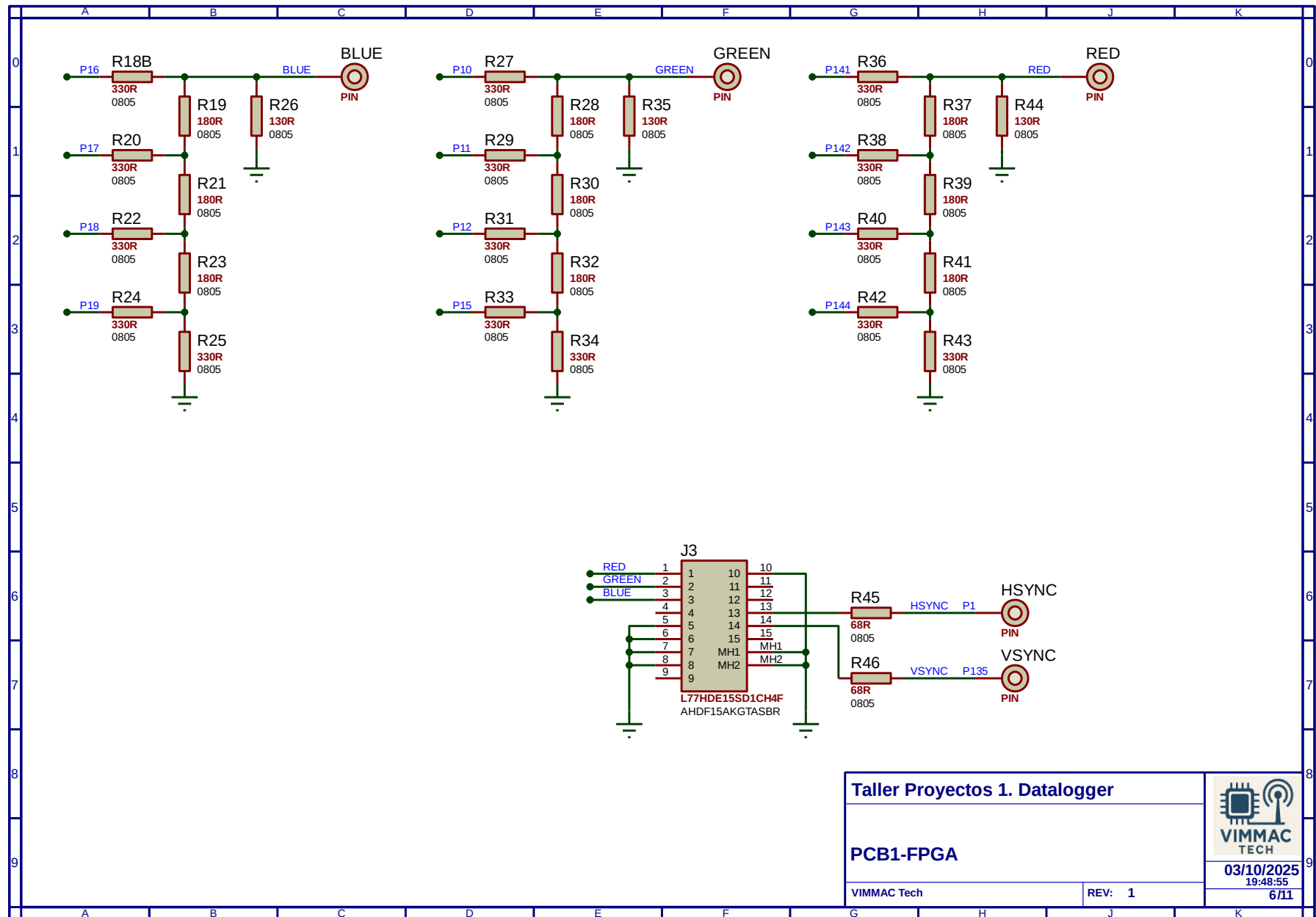


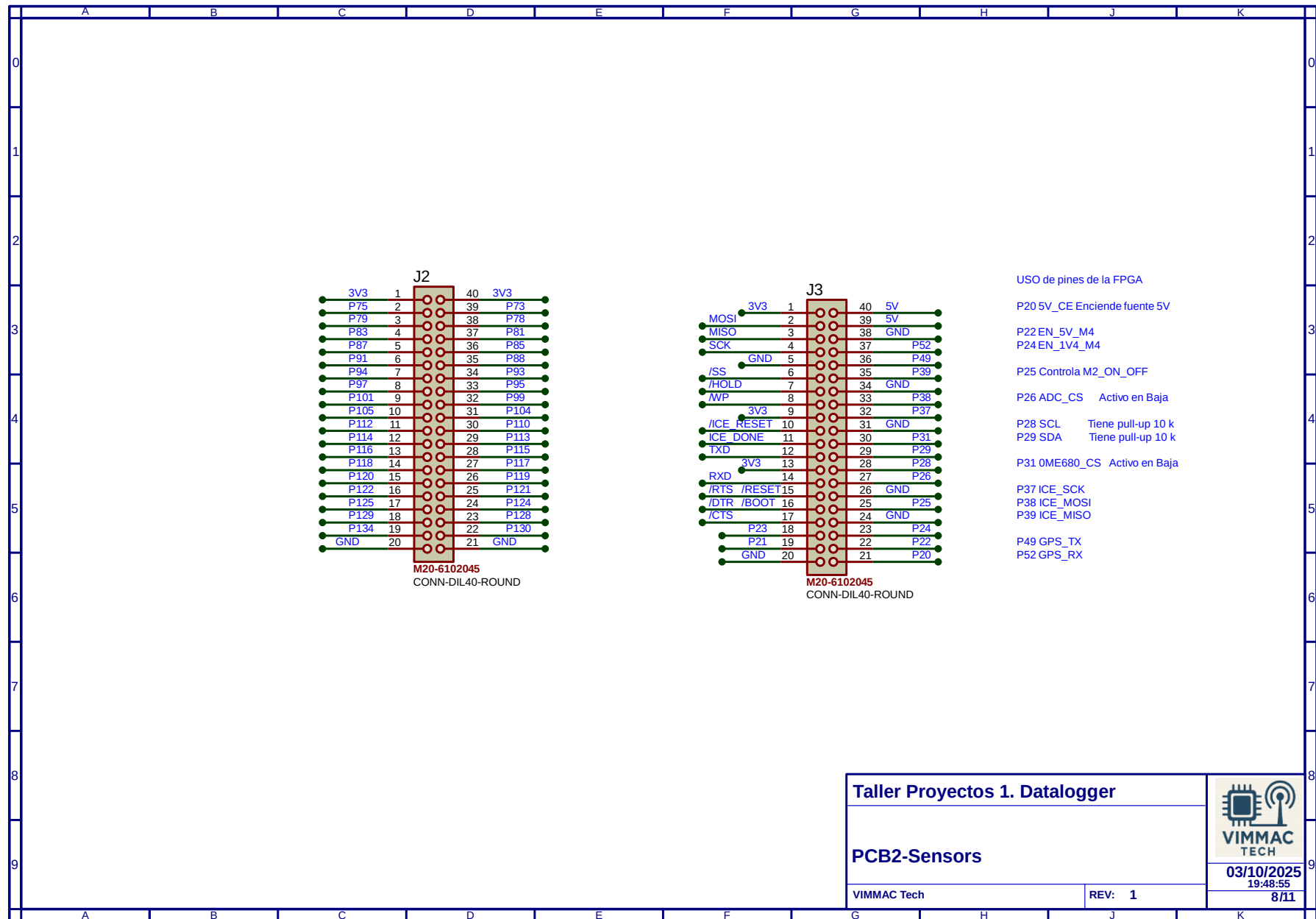
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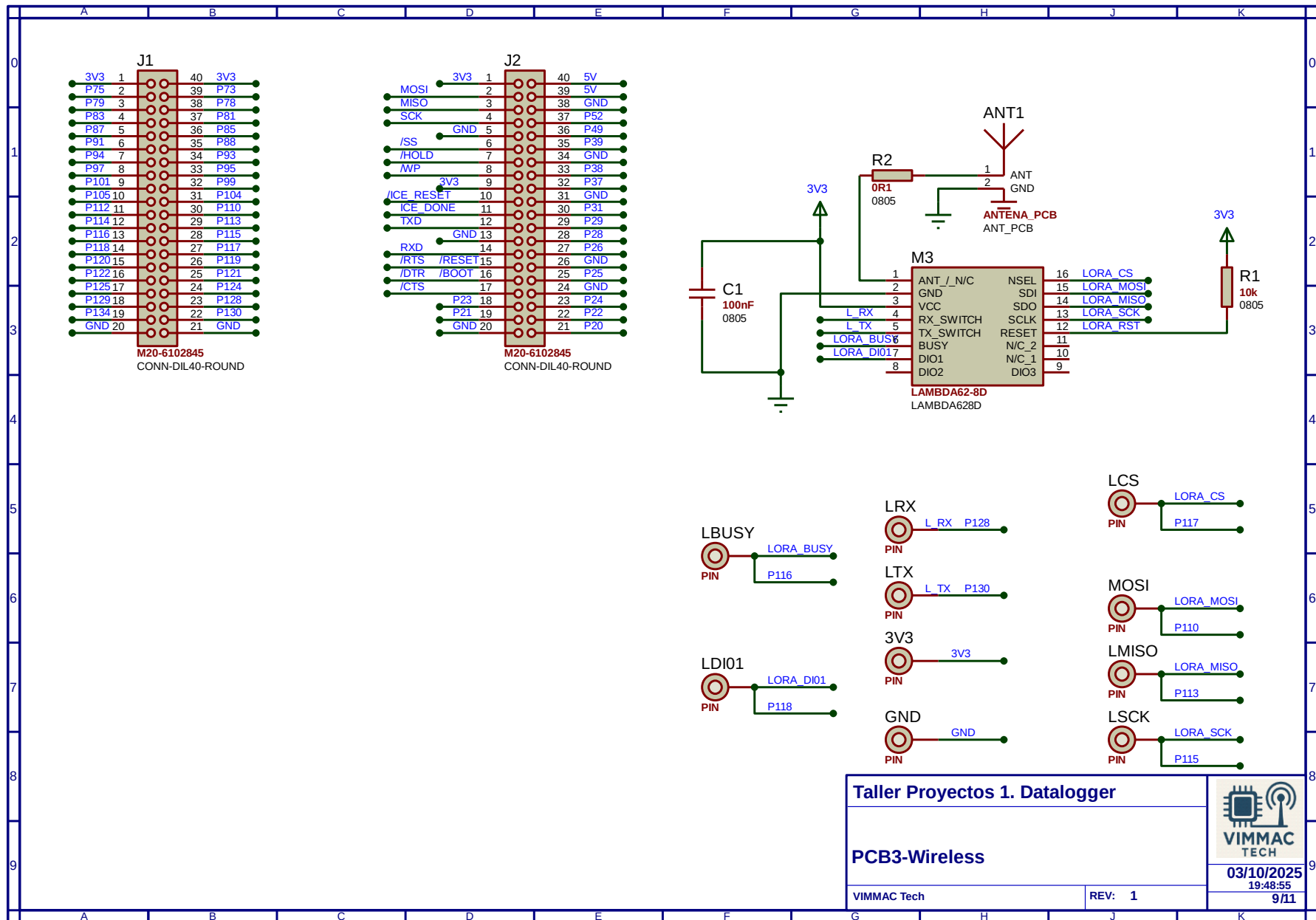




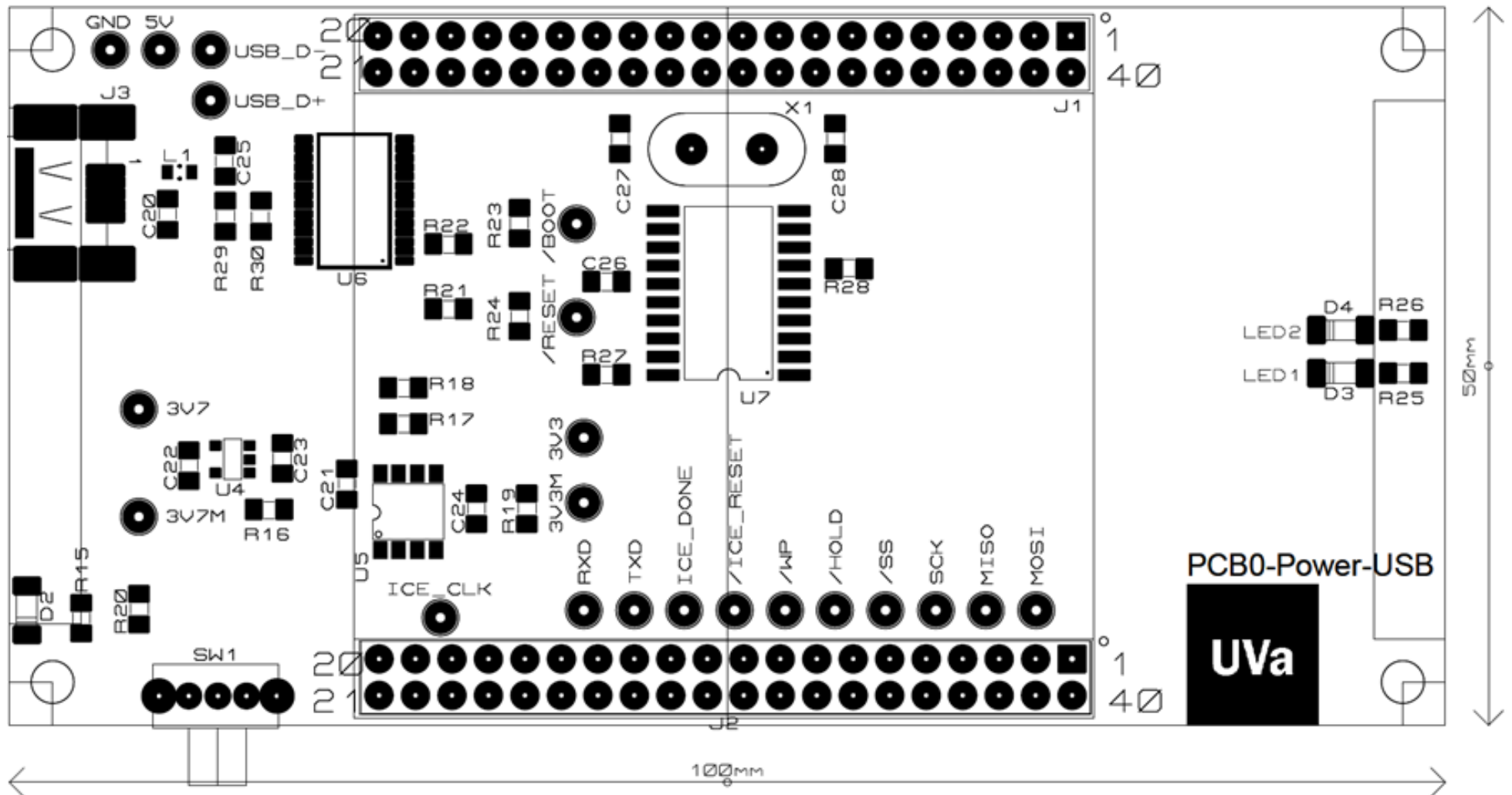


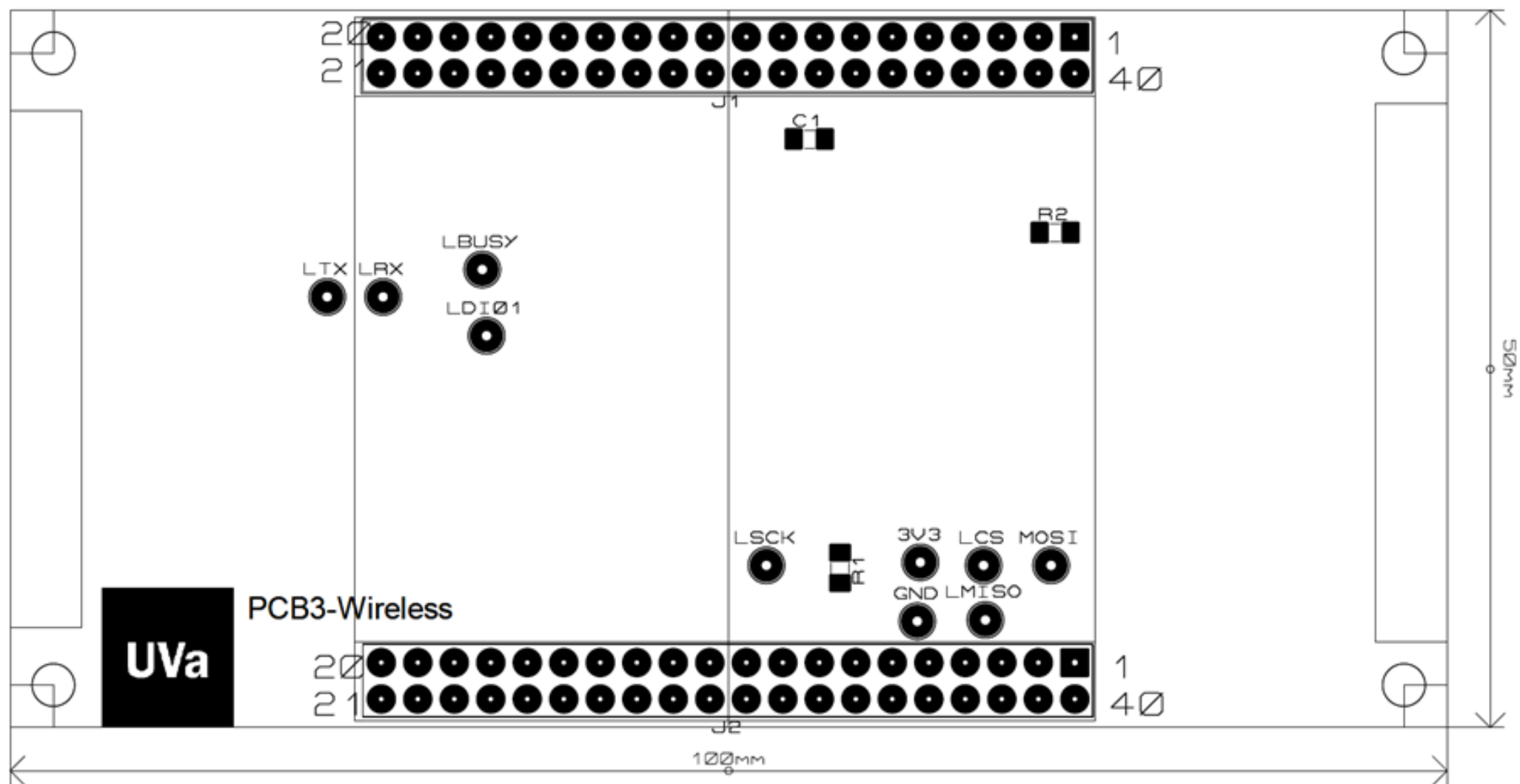






ANEXO 2: PCB





ANEXO 3: Bill Of Materials

Universidad de Valladolid
Desarrollo Práctico de Sistemas Electrónicos



Bill Of Materials for Taller Proyectos 1. Datalogger

Design Title Taller Proyectos 1. Datalogger
Author VIMMAC TECH
Document Number 1
Revision 3
Board PCB0-Power-USB
Design Created miércoles, 21 de septiembre de 2022
Design Last Modified sábado, 4 de octubre de 2025
Total Parts In Design 61

Notes

Precios sin IVA (21%), VAT not added

No incluyen gastos de envío, shipping costs not included

Algunos valores corresponden con la compra de 10 uds o más, Some values correspond to the purchase of 10 units or more.

9 Capacitors

Quantity	References	Value	PCB Package	Farnell code	Cost
2	C20,C26	100nF	0805		0,168€
4	C21,C22,C23,C24	22uF	0805		0,29325€
1	C25	10pF	0805		0,0255€
2	C27,C28	33pF	0805		0,0545€

Sub-totals:

1,64€

16 Resistors

Quantity	References	Value	PCB Package	Farnell code	Cost
1	R15	220R	0805		0,0216€
3	R16,R27,R28	2k2	0805		0,0173€
1	R17	71K5	0805		0,0364€
4	R18,R23,R24,R30	10k	0805		0,175€
2	R19,R20	0R1	0805		0,0939€
2	R21,R22	470R	0805		0,0407€
2	R25,R26	680R	0805		0,033€
1	R29	4K7	0805		0,0381€

Sub-totals:

1,18€

4 Integrated Circuits

Quantity	References	Value	PCB Package	Farnell code	Cost
1	U4	MCP73831T-2ACI/OT	SOT23-5	1332158	0,636€
1	U5	MCP1725-ADJ E/SN	S08	1834878	0,655€
1	U6	CY7C65213-28PVXI	SSOP28	2767866	1,730€
1	U7	LPC1112FD20/102	SO20W	2470837	2,56€

Sub-totals:

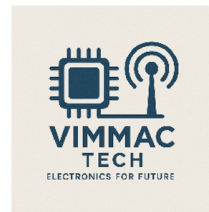
5,58€

3 Diodes

Quantity	References	Value	PCB Package	Farnell code	Cost
2	D2,D3	KA-3022SRC-4.5SF	SOD123_K	5790839	0,179€

1	D4	KA-3022SGC-4.5SF	SOD123_K	1142615	0,33€
Sub-totals:					0,69€
29 Miscellaneous					
Quantity	References	Value	PCB Package	Farnell code	Cost
21	/BOOT,/HOLD,/ICE_RESET,/RESET,/SS,/WP,3V3,3V7,3V3M,3V7M,5V,GND,ICE_CLK,ICE_DONE,MISO,MOSI,RXD,SCK,TXD,USB_D+,USB_D-	PIN	PIN		0€
1	BAT	530480310	530480310	9732918	0,260€
1	BAT1	LIPO BAT	LP735977JH	4551044	34,760€
2	J1,J2	2214S-40SG-85	CONN-DIL40-ROUND	2847248	1,86€
1	J3	USB MINI B Molex 67503-1230	MOLEX USB MINI B 67503-1230	2313554	1,140€
1	L1	MMZ1608B121C	RESC1608X84	2800823	0,0484€
1	SW1	OS102011MA1QS1	OS102011MA1QS1	2435103	0,455€
1	X1	9b-12.000MEEJ-B	HC-49S-TH	2467707	0,306€
Sub-totals:					40,69€
Totals:					49,79€

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Bill Of Materials for Taller Proyectos 1. Datalogger

Design Title Taller Proyectos 1. Datalogger
Author VIMMAC TECH
Document Number 1
Revision 3
Board PCB1-FPGA
Design Created miércoles, 21 de septiembre de 2022
Design Last Modified sábado, 4 de octubre de 2025
Total Parts In Design 94

Notes

Precios sin IVA (21%), VAT not added

No incluyen gastos de envío, shipping costs not included

Algunos valores corresponden con la compra de 10 uds o más, Some values correspond to the purchase of 10 units or more.

**Fuera de stock, referencia DigiKey: AP3216ID

20 Capacitors					
Quantity	References	Value	PCB Package	Farnell code	Cost
16	C1,C2,C3,C4,C5,C6,C7,C8,C11,C12,C13,C14,C15,C16,C17,C20	100nF	0805		0,168€
2	C9,C10	10uF	0805		0,0545 €
2	C18,C19	22uF	0805		0,29325€

Sub-totals: 3,38€

51 Resistors					
Quantity	References	Value	PCB Package	Farnell code	Cost
6	R1,R9,R10,R14,R49,R50	10k	0805		0,175€
2	R2,R47	33k	0805		0,0564€
2	R3,R4	100R	0805		0,0521€
3	R5,R6,R7	2k2	0805		0,0173€
2	R8,R48	0R1	0805		0,0939€
1	R11	12k	0805		0,420€
1	R12	4k7	0805		0,0381€
1	R13	1k5	0805		0,420€
4	R15,R16,R17,R18	680R	0805		0,033€
9	R19,R21,R23,R28,R30,R32,R37,R39,R41	180R	0805		0,0753€
15	R20,R22,R24,R25,R27,R29,R31,R33,R34,R36,R38,R40,R42,R43,R18B	330R	0805		0,270€
3	R26,R35,R44	130R	0805		0,0345€
2	R45,R46	68R	0805		0,0478€

Sub-totals: 7,44€

4 Integrated Circuits					
Quantity	References	Value	PCB Package	Farnell code	Cost
1	U1	ICE40HX4K-TQ144	QFP50P2200X2200X160-144N	3768754	9,170€
1	U2	IS25LP032D-JNLE	SO8W	2787053	0,858€
1	U3	MCP1725ADJ	SO8	1834878	0,551€

1	U4	MCSJK-3N-16-3.3-50-B	MCSJK3N16003350B	2854288	0,693€
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Sub-totals: 11,27€

5 Diodes

Quantity	References	Value	PCB Package	Farnell code	Cost
1	D1	1N4448HWS	SOD-323	3127174	0,125€
4	D2,D3,D4,D5	KA-3022SRC-4.5SF	SOD123_K	**	0,15100€

Sub-totals: 0,73€

14 Miscellaneous

Quantity	References	Value	PCB Package	Farnell code	Cost
9	1V2,1V2M,3V3,BLUE,GND,GREEN,HSYNC,RED,VSYNC	PIN	PIN		0€
2	J1,J2	CONN-DIL40	CONN-DIL40-ROUND	1569230	4,05€
1	J3	L77HDE15SD1CH4F	AHDF15AKGTASBR	2401182	1,470€
2	MB0,MB1	CONN-SIL2	CONN-SIL2	2932817	0,569€

Sub-totals: 10,71€

Totals: 33,54€

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Bill Of Materials for Taller Proyectos 1. Datalogger

Design Title Taller Proyectos 1. Datalogger
Author VIMMAC TECH
Document Number 1
Revision 3
Board PCB2-Sensors
Design Created martes, 7 de octubre de 2025
Design Last Modified martes, 7 de octubre de 2025
Total Parts In Design 55

Notes

Precios sin IVA (21%), VAT not added

No incluyen gastos de envío, shipping costs not included

Algunos valores corresponden con la compra de 10 uds o más, Some values correspond to the purchase of 10 units or more.

5 Modules					
Quantity	References	Value	PCB Package	Farnell code	Cost
1	M1	BME680_MODULE	BME680_MODULE	3932094	19,290€
1	M2	GP2Y1010AU0F	GP2Y1010AU0F	9707956	7,830€
1	M3	GPS_TEL0132	GPS_TEL0132	3769990	15,280€
1	M4	SEN0134	SEN0134	2946118	6,40€
1	M2_ON	PIN	PIN		0€
Sub-totals:					48,80€

6 Capacitors					
Quantity	References	Value	PCB Package	Farnell code	Cost
3	C1,C2,C3	1uF	0805		0,0736€
2	C4,C5	10uF	0805		0,0545€
1	C6	EEEFK1V221P		9695877	0,390€
Sub-totals:					0,72€

11 Resistors					
Quantity	References	Value	PCB Package	Farnell code	Cost
6	R1,R2,R5,R6,R7,R11	10k	0805		0,175€
1	R3	6K8	0805		0,366€
1	R4	1K2	0805		0,366€
1	R8	820R	0805		0,0887€
1	R9	470R	0805		0,0407€
1	R10	150R	0805		0,0585€
Sub-totals:					1,97€

3 Integrated Circuits					
Quantity	References	Value	PCB Package	Farnell code	Cost
1	U1	MS8607-02BA01	QFN_5X3_NEG	2748850	7,980€
1	U2	MCP3004-I/SL	SO14	1852016	2,190€
1	U3	XC9141B50DMR-G	SOT23-5	3535653	0,764€
Sub-totals:					10,93€

4 Transistors					
Quantity	References	Value	PCB Package	Farnell code	Cost
3	Q1,Q3,Q4	MMBT3904-7-F	SOT95P230X109-3	1773602	0,110€
1	Q2	NDS332P	SOT23	1471069	0,527€
Sub-totals:					0,86€
26 Miscellaneous					
Quantity	References	Value	PCB Package	Farnell code	Cost
21	3V3,5V_CE,5V_M4,5V_UP,ADCCS,BMECS,CH0,CH1,CH2,CH3,EN1V4,EN5V,GND,GPS_RX,GPS_TX,ICEMISO,ICEMOSI,ICESCK,P25,SCL,SDA	PIN	PIN		0€
1	J1	S6BZRSM4ATFLFSN	JST_S6BZRSM4ATFLFSN	1830924	0,532€
2	J2,J3	M20-6102045	CONN-DIL40-ROUND	1569230	4,05€
1	J4	ZHR-6	JST_ZHR-6	3357570	0,0665€
1	L1	LQH5BPNR47NT0L		2219309	0,265€
Sub-totals:					8,96€
Totals:					72,24€

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Bill Of Materials for Taller Proyectos 1. Datalogger

Design Title Taller Proyectos 1. Datalogger
Author VIMMAC TECH
Document Number 1
Revision 3
Board PCB3-Wireless
Design Created martes, 7 de octubre de 2025
Design Last Modified martes, 7 de octubre de 2025
Total Parts In Design 20

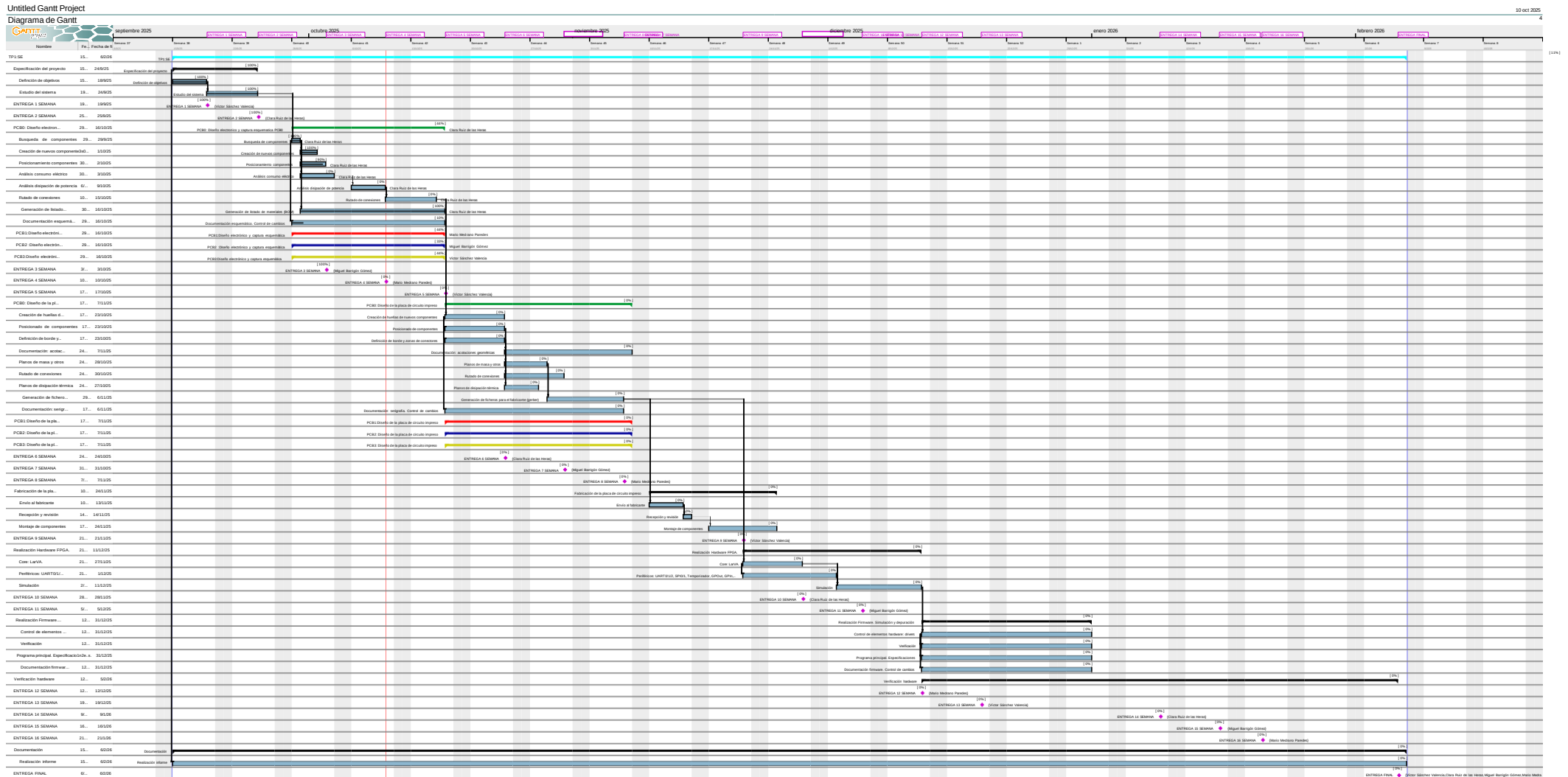
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Precios sin IVA (21%), VAT not added
No incluyen gastos de envío, shipping costs not included
Algunos valores corresponden con la compra de 10 uds o más, Some values correspond to the purchase of 10 units or more.
**Fuera de stock, referencia DigiKey: AP3216ID

1 Modules					
Quantity	References	Value	PCB Package	Farnell code	Cost
1	M3	LAMBDA62-8D	LAMBDA628D	2988568	14,44€
Sub-totals:					14,44€
3 Capacitors					
Quantity	References	Value	PCB Package	Farnell code	Cost
1	C1	100nF	0805		0,168€
2	C2,C3		0805		
Sub-totals:					0,17€
2 Resistors					
Quantity	References	Value	PCB Package	Farnell code	Cost
1	R1	10k	0805		0,175€
1	R2	0R1	0805		0,0939€
Sub-totals:					0,27€
14 Miscellaneous					
Quantity	References	Value	PCB Package	Farnell code	Cost
10	3V3,GND,LBUSY,LCS,LDI01,LMISO,LRX,LSCK,LTX,MOSI	PIN	PIN		0€
1	ANT1		NULL		
2	J1,J2	M20-6102845	CONN-DIL40-ROUND	1569230	4,05€
1	L1		INDC3225X135		
Sub-totals:					8,10€
Totals:					22,98€

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ANEXO 4: DIAGRAMA GANTT



ANEXO 5: DIAGRAMA PERT

